Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – WEEKLY TEST 06
Chapter 1: Real Numbers

Time: 45 Minutes
MM: 20

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer (VSA) question of 2 marks.
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 1 Case Study based question of 4 marks with internal subparts.
6. Use of calculators is strictly prohibited. All mathematical steps must be clearly shown.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. The sum of the exponents of prime factors in the prime factorisation of 196 is: [1]
(a) 1 (b) 2 (c) 4 (d) 6

Q2. The sum of the exponents of prime factors in the prime factorisation of 144 is: [1]
(a) 4 (b) 5 (c) 6 (d) 7

Q3. If two positive integers m and n are expressed as $m = p^2q^3$ and $n = p^3q$, where p and q are primes, then $\text{HCF}(m, n) \times \text{LCM}(m, n)$ is: [1]
(a) p^5q^4 (b) p^6q^3 (c) p^5q^3 (d) p^6q^4

Q4. If p_1 and p_2 are two odd prime numbers such that $p_1 > p_2$, then $p_1^2 - p_2^2$ is always: [1]
(a) an odd integer (b) an even integer
(c) an odd prime (d) a fractional number

Q5. **Assertion (A):** $2 + \sqrt{3}$ is an irrational number.
Reason (R): The sum of a rational number and an irrational number is always an irrational number. [1]
(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (1 × 2 = 2 Marks)

Q6. Find the prime factorisation of the number 32760 and write it in exponential form. [2]

SECTION C: Short Answer Type (3 × 3 = 9 Marks)

Q7. Find the largest positive integer that will divide 398, 436, and 542 leaving remainders 7, 11, and 15 respectively. [3]

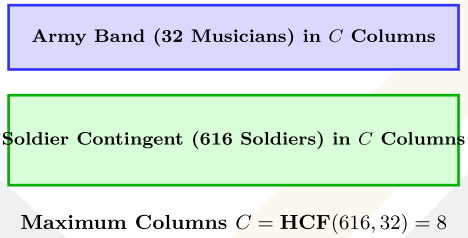
Q8. Find the least number which when divided by 20, 25, 35, and 40 leaves remainders 14, 19, 29, and 34 respectively. [3]

Q9. Find the smallest number which leaves remainder 8 and 12 when divided by 28 and 32 respectively. [3]

SECTION D: Case Study Based Question (1 × 4 = 4 Marks)

Q10. CASE STUDY: CASE STUDY 06: REPUBLIC DAY ARMY CONTINGENT PARADE MARCHING [4 Marks]

During the annual Republic Day parade rehearsals at Kartavya Path, an army contingent of 616 soldiers is scheduled to march behind an army band of 32 musicians. The parade drill commander mandates that the two groups must march in the exact same number of columns, with each column possessing a uniform alignment and discipline.



(i) Write the prime factorisations of 616 and 32. [1]

(ii) Find the maximum number of columns in which the band and contingent can march. [2]

(iii) How many contingent soldiers and band members will march in each column respectively? [1]

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